Supplementary Results for KDD 2026 Rebuttal

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Overview

This document provides supplementary empirical evidence, mechanism diagnostics, and quantitative benchmarks to support the primary rebuttal for our KDD 2026 submission. To address the reviewers' inquiries with rigorous transparency, we have systematically expanded our evaluation protocols. Specifically, this document details: (1) real-world interpretability validation under strict faithfulness metrics (Reviewers gq6s, gMhY); (2) comprehensive internal utilization and gating diagnostics to verify healthy expert specialization (Reviewer Hjgq); (3) unified, explanation-aware throughput benchmarks at extreme scales (Reviewer gMhY); and (4) structural sensitivity analyses alongside reproducible tuning heuristics (Reviewer ncgk). Detailed tabular records corresponding to all visual diagnostics are compiled in Section 5.

1 Additional Experiment Results to Reviewer gq6s

Interpretability Validation (W1, W5)

We evaluate real-data interpretability using the `I(20)` and `I(100)` metrics. Concretely, all explainers are evaluated under the same ranking, masking, and retraining pipeline with a shared evaluator: `I(20)` reports predictive performance after retaining the top-ranked 20% evidence budget, whereas `I(100)` reports the corresponding full-budget score. Higher scores combined with lower average ranks indicate that the highlighted temporal evidence effectively preserves predictive signals.

Table 1 summarizes the comprehensive quantitative faithfulness results under a fixed masking and retraining protocol. Under the stringent `I(20)` threshold, AnchorMoE achieves 0.6419, exceeding both the strongest time-series specific baseline (**FIT**, 0.5760) and the generic baseline (**IG**, 0.6189). Under `I(100)`, AnchorMoE reaches 0.8620, outperforming **Dynamask** (0.8284). These results confirm that the temporal evidence identified by our model remains highly predictive on real data. For reference, the compared generic explainers are **Integrated Gradients**, **DeepLift**, and **Grad-CAM**, while the time-series-specific baselines follow **Dynamask** [1] and the original **FIT** [2] formulation.

2 Additional Experiment Results to Reviewer Hjgq

Principal Explanation (W1, W3, W4, W5)

To address concerns regarding the function of gradients within our framework, we isolate their role during training. Referring back to the global evaluation in Table 1, AnchorMoE (w/o grad) still comprehensively exceeds the **Integrated Gradients** baseline on the mean `I(100)` score (0.8667 vs. 0.8384). However, the standard gradient-calibrated AnchorMoE model maintains an advantage under the more restrictive `I(20)` threshold. This performance pattern validates our core design premise: gradient magnitude functions as an instance-level calibration signal during training, rather than the final explanatory output at inference.

Internal Mechanism (W2, W7, W8, W9)

To address concerns regarding potential expert collapse, we comprehensively tracked internal utilization statistics. Specifically, `Eff. Experts` is computed as the inverse Simpson index of the token-to-expert routing distribution. `Redundancy` is evaluated as a posterior-anchor redundancy proxy, measured by the off-diagonal squared mean of the cosine similarity Gram matrix formed by dynamic posterior anchors. Representation collapse would manifest as a critically low active rate, near-unit max share, and effective experts approaching 1.0, whereas healthy specialization preserves non-trivial utilization without enforcing a uniform regime.

The visual distributions in Figure 1 directly demonstrate that the routing concentration and confidence sparsity dynamically adapt to dataset characteristics rather than collapsing into a single pattern. For detailed dataset-specific tabular records concerning expert utilization and confidence sparsity, please refer to Tables 4 and 5 in Section 5.

To systematically evaluate the sequential stages of the interpretable decision pipeline, Table 2 is structured into discrete diagnostic blocks. Specifically, B1 examines query formulation, B2 investigates expert specialization, and B3 assesses selective evidence binding. Within the MVSE module (B1), ablating the sample-level relevance cue yields a severe accuracy drop (0.9686 $\longrightarrow$ 0.7084). This confirms that the router depends on sample-conditioned signals for spatial disambiguation, rather than relying on isolated local views. Conversely, omitting the spectral branch introduces a measured performance decay (0.9686 $\longrightarrow$ 0.9462), suggesting its role as a phase-robust structural prior that stabilizes, rather than dominates, the discriminative process. For the routing mechanism (B2), the critical indicator lies in semantic geometry. Without DPAR, inter-anchor redundancy escalates from 0.3488 to 0.7213, while the effective number of experts remains relatively static (2.4280 vs. 2.4563). Empirical data indicates that the introduction of DPAR effectively mitigates semantic redundancy among experts, and its geometric constraints play a critical role in preserving effective expert activation, moving beyond conventional traffic-balancing heuristics. Finally, in the context of confidence-weighted binding (B3), removing UADB drives the mean κ to 1.0000 and penalizes early-stage faithfulness ($I(20)$ decreases from 0.8953 to 0.8819), yet leaves the expanded-budget metric $I(100)$ largely unaffected. This metric divergence demonstrates that while the baseline routing continues to capture candidate evidence, the model loses the capacity to prioritize critical patches. Consequently, UADB dictates selective binding to explicitly sharpen evidence concentration, rather than merely preserving aggregate evidence.

Table 1: Real-data faithfulness evaluation. Higher is better for `I(20)` and `I(100)`, while lower is better for rank. The best and second-best results in each row are highlighted.

Dataset / Stat.	Metric	AnchorMoE	AnchorMoE (w/o grad)	IG	DeepLift	Grad-CAM	Random	Dynamask	FIT
Overall Mean	I(20)	**0.6419**	**0.6338**	0.6189	0.6168	0.5891	0.5301	0.5390	0.5760
	I(100)	**0.8620**	**0.8667**	0.8384	0.8363	0.8278	0.8035	0.8284	0.8264
Overall Rank	I(20)	**2.86**	3.43	**3.14**	3.86	4.71	6.29	6.00	5.71
	I(100)	**2.57**	**2.86**	3.71	4.43	5.86	6.29	4.57	5.71
ArticularyWordRec.	I(20)	0.6554	0.6625	**0.6808**	**0.6749**	0.5832	**0.6749**	0.5757	0.6361
	I(100)	0.8772	0.8810	0.8812	**0.8963**	0.8599	**0.8938**	0.8812	0.8725
BasicMotions	I(20)	**0.5750**	**0.5938**	0.4625	0.5094	0.5521	0.4583	0.5083	0.4531
	I(100)	**0.8175**	**0.8675**	0.7354	0.7306	0.7792	0.7388	0.8092	0.7819
Cricket	I(20)	0.6597	0.6111	**0.6985**	**0.6997**	0.6140	0.6198	0.6024	0.6869
	I(100)	0.8993	0.8819	**0.9041**	**0.9078**	0.8703	0.8443	0.8760	0.8892
FordA	I(20)	0.6661	**0.6821**	0.6301	0.6412	0.6469	0.4620	**0.6920**	0.5631
	I(100)	0.8678	**0.8766**	0.8401	0.8287	0.8541	0.7505	**0.8800**	0.8061
NATOPS	I(20)	0.6764	0.6708	**0.6792**	**0.6769**	0.6069	0.5315	0.4843	0.5486
	I(100)	**0.8883**	**0.8808**	0.8744	0.8671	0.8517	0.8500	0.7591	0.8252
RacketSports	I(20)	**0.5576**	0.5206	0.4729	0.4482	**0.5414**	0.4238	0.4833	0.4512
	I(100)	**0.7566**	**0.7541**	0.7054	0.7053	0.7194	0.6762	0.7278	0.6910
SpokenArabicDigits	I(20)	**0.7030**	0.6959	**0.7081**	0.6675	0.5790	0.5405	0.4271	0.6928
	I(100)	**0.9274**	0.9251	**0.9283**	0.9180	0.8600	0.8711	0.8658	0.9191

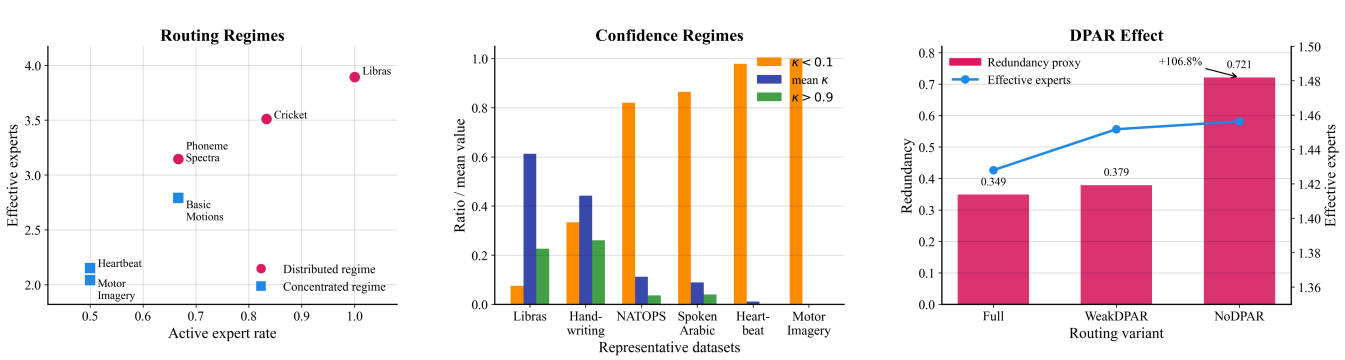

Figure 1: Mechanism diagnostics across 29 datasets, highlighting dataset-dependent utilization and confidence behavior.

Table 2: Comprehensive internal-design ablation and mechanism diagnostics. Experiments are grouped into discrete blocks (B1: MVSE, B2: DPAR, B3: UADB). Metrics marked with '–' are invariant or non-applicable to the specific diagnostic block. Because diagnostic subsets vary across blocks, absolute scores differ and are not directly comparable to the 18-dataset overall averages in the main paper.

Block	Variant	Mean Acc	I(20)	I(100)	Eff. Experts	Redundancy	Mean κ
B1	**Full**	0.9686	0.9250	0.9348	–	–	–
	NoRelevance	0.7084	0.8478	0.9348	–	–	–
	NoSpectral	0.9462	0.9268	0.9348	–	–	–
B2	**Full**	0.9686	0.9250	0.9348	2.4280	0.3488	0.5319
	NoDPAR	0.9727	0.9580	0.9348	2.4563	0.7213	0.5600
	WeakDPAR	0.9734	0.9518	0.9348	2.4518	0.3788	0.5306
B3	**Full**	0.9509	0.8953	0.9081	–	–	0.3833
	NoConfidence	0.9477	0.8819	0.9078	–	–	1.0000

3 Additional Experiment Results to Reviewer gMhY

Scalability Analysis & Runtime (W1)

To comprehensively address scalability concerns, we re-evaluate the framework focusing on the end-to-end throughput required for transparent decision-making. We rerun the benchmark under a unified synthetic 6-class multivariate-classification protocol. AnchorMoE extracts its intrinsic explanation seamlessly during the routed forward pass, whereas every baseline (including **MiniRocket**) is charged with a fixed-budget temporal occlusion procedure using 10 windows. Each run is capped at 400 seconds.

Figure 2 visualizes the results of this extended benchmark. Because AnchorMoE bypasses the computational tax associated with

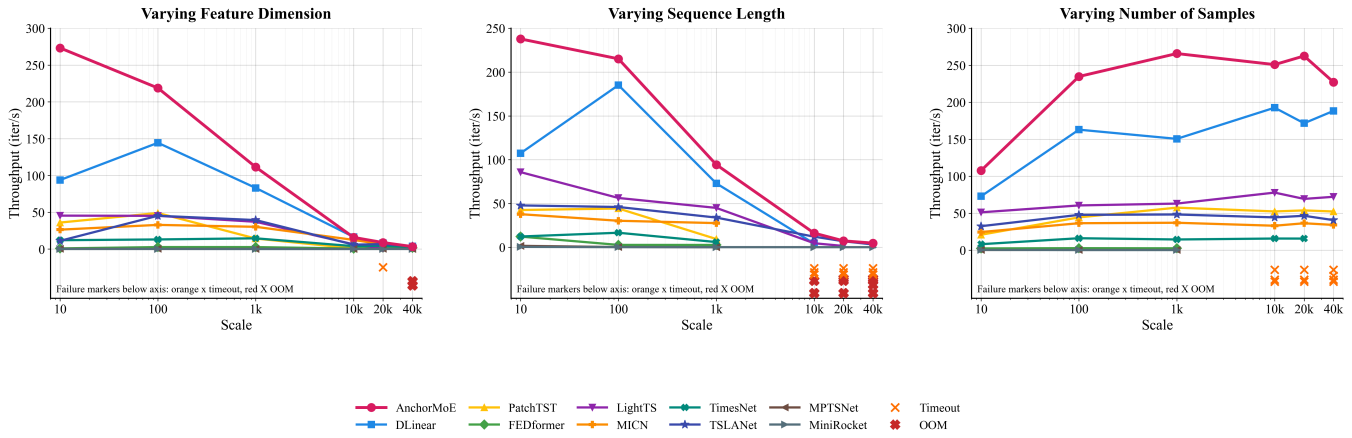

Figure 2: Explanation-aware throughput trends under the unified protocol. Solid lines denote completed runs, while markers below the axis indicate timeout or out-of-memory failures.

post-hoc attribution, it successfully outperforms the strongest completed baseline at each point in 16 of 18 evaluated configurations. It demonstrates particular dominance along the sequence length and sample size axes, winning all test points. For instance, at the extreme 40,000 sequence length mark, AnchorMoE achieves a 1.48× speedup compared to **TSLANet**. Crucially, while multiple baselines trigger timeout (TO) or out-of-memory (OOM) failures at extreme scales, our framework maintains robust execution. For a comprehensive perspective on pure inference without intrinsic explanation extraction, our theoretical analysis (Section 3.5) shows AnchorMoE operates with a linear complexity of $O(LKD)$, structurally bypassing the quadratic computational bottleneck of standard attention utilized by pure classifiers. Comprehensive quantitative throughput results are provided in Table 3 in Section 5.

Interpretability on Real Data (W2)

As evaluated in Section 1, we employ the strict `I(20)` and `I(100)` metrics to measure predictive temporal localization. The corresponding visual and tabular results confirm that our framework consistently preserves structural integrity and maintains a compact temporal focus on real-world datasets.

4 Additional Experiment Results to Reviewer ncgk

Tuning Complexity & Anchor Determination (W1, W3)

To assess tuning sensitivity, we track the model response across anchor diversity, spectral weight, group count, and confidence regularization. The empirical evidence counters the assumption of uniform brittleness. While structural factors such as anchor diversity present relatively stable operating regions, the confidence regularization weight requires conservative tuning. Specifically, on the more difficult NATOPS dataset, aggressive confidence regularization drives accuracy sharply from 0.9500 to near chance-level (0.1667), as reflected by the `conf_lambda` profile summarized in Figure 3 (with detailed bounds provided in Table 6, Section 5).

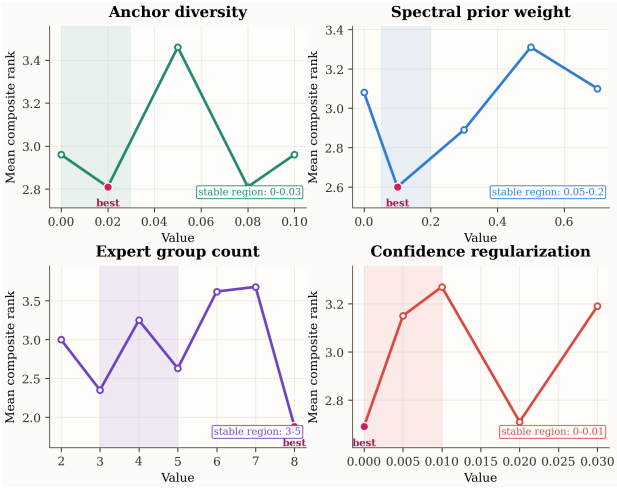

Figure 3: Performance sensitivity curves across primary structural hyperparameters.

We attribute this degradation to an evidence-gating collapse rather than routing collapse. Because the confidence loss explicitly regulates the relative distribution of the confidence gate against a self-bootstrapped gradient target without constraining the absolute confidence mass, an aggressive penalty on noisy early-stage gradients forces the network to severely suppress the absolute confidence magnitude across most patches. This extreme filtering diminishes the forward evidence flow, scaling down backpropagated gradients and ultimately inducing a gradient starvation loop. Removing the confidence constraint averts this collapse but induces a precision-recall trade-off, where strict top-budget faithfulness (`I(20)`) drops while the expanded-budget score (`I(100)`) improves. To mitigate tuning brittleness, we propose a reproducible heuristic for determining the optimal number of groups: selecting the minimal count that achieves validation plateau. As supported by Table 7 in Section 5, this optimum consistently falls within a predictable range of 3 to 5 groups.

5 Detailed Quantitative Results

This section provides the supplementary tabular data supporting the qualitative and visual trends discussed in the main text.

Table 3: Throughput comparison (iter/s) under the unified protocol. The best and second-best throughputs in each row are highlighted. 'TO' and 'OOM' denote timeout and out-of-memory failures.

Axis	Scale	AnchorMoE	PatchTST	FED former	DLinear	LightTS	MICN	TimesNet	TSLANet	MPTSNet	Mini Rocket
Feature-dimension sweep											
Feature Dim.	10	**273.303**	36.039	0.920	**93.836**	45.451	26.393	12.059	11.181	0.278	0.120
	100	**218.865**	49.023	2.709	**144.414**	45.171	32.882	13.089	45.296	0.261	0.188
	1000	**111.366**	14.323	2.697	**83.065**	37.089	30.262	14.627	39.410	0.243	0.072
	10k	**15.882**	0.982	0.693	**16.532**	6.510	12.483	3.968	5.977	0.081	0.056
	20k	**8.879**	TO	1.940	**8.324**	0.892	6.232	4.528	6.817	0.268	0.156
	40k	3.374	OOM	1.636	**3.898**	OOM	3.459	2.778	**3.853**	0.252	0.148
Sequence-length sweep											
Sequence Len.	10	**237.871**	42.578	12.226	**107.408**	85.805	37.816	12.334	47.824	1.672	0.472
	100	**215.164**	44.297	2.737	**185.154**	56.326	30.299	16.623	46.015	0.257	0.188
	1000	**94.381**	9.466	2.683	**73.020**	44.992	27.633	5.985	33.941	0.029	0.167
	10k	**16.198**	OOM	TO	4.275	4.655	TO	TO	**12.136**	OOM	0.156
	20k	**7.347**	OOM	TO	OOM	1.516	TO	TO	**6.896**	OOM	0.148
	40k	**4.804**	OOM	TO	OOM	OOM	TO	TO	**3.249**	OOM	0.130
Number-of-samples sweep											
Num. Samples	10	**107.708**	20.893	2.390	**72.935**	51.193	24.553	8.103	32.451	0.283	0.398
	100	**234.845**	44.372	2.702	**162.943**	60.522	36.469	16.167	47.770	0.239	0.187
	1000	**265.869**	57.499	2.539	**150.518**	62.998	37.092	14.474	48.258	0.275	0.189
	10k	**250.911**	52.411	TO	**192.657**	77.862	33.067	15.747	44.341	TO	TO
	20k	**262.604**	53.769	TO	**171.634**	69.332	36.402	15.688	46.513	TO	TO
	40k	**227.131**	52.324	TO	**188.349**	72.191	34.161	TO	40.647	TO	TO

Table 4: Dataset-specific expert utilization behavior corresponding to the visual mechanism diagnostics.

Dataset	Active Rate	Max Share	Effective Experts
Libras	1.0000	0.2932	3.8933
Cricket	0.8333	0.3194	3.5120
PhonemeSpectra	0.6667	0.3405	3.1461
BasicMotions	0.6667	0.3856	2.7905
Heartbeat	0.5000	0.4600	2.1503
MotorImagery	0.5000	0.4800	2.0412

Table 5: Dataset-specific κ statistics, detailing the sparsity of the confidence gating mechanism.

Dataset	Mean κ	$\kappa < 0.1$	$\kappa > 0.9$
Libras	0.61276	0.07514	0.22591
Handwriting	0.44180	0.33335	0.26026
NATOPS	0.11251	0.81994	0.03575
SpokenArabicDigits	0.08893	0.86469	0.03999
Heartbeat	0.01114	0.97854	0.00000
MotorImagery	0.00007	0.99996	0.00000

Table 6: Sensitivity analysis summary detailing quantitative performance boundaries under key structural hyperparameters.

Parameter	Scanned values	Acc. Range
Anchor Div.	0, 0.02, 0.05, 0.08, 0.1	0.9268–0.9332
Spectral Wt.	0, 0.1, 0.3, 0.5, 0.7	0.9281–0.9374
Num. Groups	2, 3, 4, 5, 6, 7, 8	0.9166–0.9722
Conf. λ	0, 0.005, 0.01, 0.02, 0.03	0.7976–0.9314

Table 7: Dataset-specific optimal group counts derived from the validation plateau heuristic.

Dataset	Best num_groups	Acc	F1	AOPCAcc	IoU
ArticularyWord Recognition	3	0.9733	0.9733	0.7898	0.2820
Cricket	5	0.9722	0.9720	0.7000	0.2796
FordA	3	0.9356	0.9356	0.3461	0.2499
Heartbeat	4	0.7707	0.7674	0.4898	0.2769
NATOPS	4	0.9833	0.9833	0.6578	0.3750
SpokenArabic Digits	3	0.9900	0.9900	0.7773	0.3007

Risk of Over-filtering & Safety-Critical Trade-offs (W2)

Building on the real-data evaluations, the B3 block of Table 2 (Section 2) investigates the risk of over-filtering by ablating the intrinsic confidence score entirely. Without this self-assessment capability, the model reverts to a uniform gating regime ($\kappa = 1.0$). Consequently, as background noise dilutes genuine discriminative signals, the strict `I(20)` score degrades from 0.8953 to 0.8819. This structural degradation validates our core design premise: while the routing mechanism governs spatial feature allocation, the confidence score provides an indispensable filtering layer to suppress weak evidence.

References

[1] Jonathan Crabbé and Mihaela Van Der Schaar. 2021. Explaining Time Series Predictions with Dynamic Masks. In *Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139).* PMLR, Virtual, 2166–2177. https://proceedings.mlr.press/v139/crabbe21a.html

[2] Sana Tonekaboni, Shalmali Joshi, Kieran Campbell, David K. Duvenaud, and Anna Goldenberg. 2020. What went wrong and when? Instance-wise feature importance for time-series black-box models. In *Advances in Neural Information Processing Systems 33.* Curran Associates, Inc., Vancouver, Canada. https://proceedings.neurips.cc/paper_files/paper/2020/hash/08fa43588c2571ade19bc0fa5936e028-Abstract.html